

CRADLE: Transient Security and Dependability Analysis of Crypto-Ransomware Kill-Chains in MEC Micro-Datacenters under Footprint-Aware Detection and Backup-Restore Defense

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Abstract—Multi-access edge computing places a small number of virtualized servers beside base stations, and such micro-datacenters face crypto-ransomware campaigns of growing severity. A ransomware incident differs from a classical breach because locked servers remain restorable from immutable backups. The true cost of an incident becomes the transient loss of confidentiality, service availability and data recency accrued during the race between the attacker kill-chain and the defender response. The present work develops CRADLE, a continuous-time Markov reward model drawn as a stochastic reward net, of an ordered compromise, exfiltration, encryption and lock kill-chain on a micro-datacenter. The defense combines an endpoint monitor whose detection hazard grows with the encryption and exfiltration footprint and an isolate-and-restore response with explicit recovery-point and recovery-time semantics. The model delivers transient state probabilities, accumulated-reward security losses, first-passage detection and resolution distributions, and closed per-incident expectations from the fundamental matrix. A proof establishes a structural departure, under which footprint-aware detection breaks the memoryless recovery law of prior uniform-recovery models. A defender co-design procedure tunes detection sensitivity and backup cadence jointly along a security-versus-spend Pareto frontier. An independent Gillespie simulation validates every analytic result, and the numerical evidence gives actionable guidance for edge ransomware resilience.

Index Terms—Multi-access edge computing, ransomware, kill-chain, stochastic reward nets, continuous-time Markov chains, transient analysis, dependability, security, backup and restore, performability.

I. INTRODUCTION

MULTI-ACCESS edge computing (MEC) pushes computation to micro-datacenters (MEDCs) of a few virtualized servers placed beside base stations, and the resulting proximity delivers the low latency demanded by immersive, industrial and vehicular applications [1], [2]. Such proximity creates a substantial attack surface at the same time. An MEDC occupies a physically and administratively exposed location, hosts multi-tenant virtual machines and containers, and forms a natural staging point for adversaries seeking the data and the control flows traversing the edge [3]. The

dominant monetized threat against such edge fleets today is crypto-ransomware. An adversary gains an initial foothold, moves laterally between servers, exfiltrates data for double-extortion leverage, and then encrypts data at rest to deny service until a ransom is paid [4], [5], [6].

Quantitative stochastic models have proven invaluable for reasoning about the security and the dependability of computing systems before deployment. The MEDC-under-attack model of Liu *et al.* [13] is the closest antecedent. The antecedent analyses a lateral-movement campaign on an MEDC as an absorbing continuous-time Markov chain (CTMC) and computes transient infection, exfiltration and loss metrics. Two simplifications of the antecedent model matter a great deal for ransomware. First, recovery is a single uniform rate to an absorbing state in which defensive mechanisms have been deployed, and the rate carries no dependence on the progress of the attack, which makes the recovery probability exactly memoryless. Detection in practice is driven by the noise of the attacker because encryption produces a high-entropy input-output storm and mass data egress, and an endpoint monitor observes such activity precisely because the campaign is advancing. Second, the antecedent carries no notion of data-at-rest denial and of the reversal of such denial. Encryption is absent as a recoverable service-denial state, and no backup-and-restore layer supplies recovery-point or recovery-time semantics, which are exactly the levers under defender control.

The present work closes the gap left by the antecedent model. We propose CRADLE, a transient Markov reward model expressed as a stochastic reward net (SRN), of a crypto-ransomware double-extortion kill-chain on an MEDC, defended by a footprint-aware endpoint detection-and-response (EDR) monitor and by an immutable-backup isolate-and-restore response. A ransomware incident inside the CRADLE model remains survivable by construction. Every trajectory eventually reaches a restored condition, and the object of study becomes the transient security and dependability loss in confidentiality, service availability, data recency and downtime accumulated while the kill-chain races the response.

The main contributions of the present study appear in the following list.

- An ordered double-extortion ransomware SRN (Section III) whose per-server kill-chain runs from vulnerable to compromised, to exfiltrating, to encrypting and

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All models, solvers, the simulator, and the scripts regenerating every figure, table and in-text number are openly available at <https://github.com/anhnt2407/273-CRADLE-Transient-Ransomware-Kill-Chain-Detection-Backup-Restore-MEC-Micro-Datacenter-SRN>.

to locked, with lateral movement scaling in the active footprint, and in which encryption becomes a recoverable service-denial state rather than an absorbing breach.

- A footprint-aware detection hazard keyed on the encryption and exfiltration input-output footprint, coupled to an isolate-and-restore response with explicit recovery-point and recovery-time semantics and with imperfect restore coverage (Section III). We prove the coupling breaks the memoryless recovery law of uniform-recovery models, and the resolution curve takes an S shape instead of an exponential shape.
- Exact transient and per-incident measures (Section IV), namely transient probabilities and accumulated-reward losses through matrix exponentials, first-passage detection and resolution distributions, and closed per-incident expectations through the fundamental matrix.
- A defender co-design algorithm (Section IV) optimizing detection sensitivity and backup cadence jointly to minimize expected total cost, tracing the security-versus-spend Pareto frontier and quantifying the penalty of a detection-blind backup schedule.
- A dual analytic and simulation validation plus a numerical study (Section V) surfacing actionable design guidance, including the counterintuitive result under which sharper detection lets a defender back up less often.

II. RELATED WORK

A. Stochastic dependability and security modelling

Markov reward models, stochastic Petri and reward nets, and semi-Markov processes are the standard vehicles for quantifying availability, reliability and security of computing systems [14], [15], [17], [18]. Meyer introduced performability as the joint treatment of performance and dependability [16], and Nicol *et al.* extended model-based evaluation from dependability toward quantitative security [17]. The dependability and security taxonomy of Avižienis *et al.* [20] defines integrity and availability as distinct attributes, and encryption-based denial affects both attributes at once through a fault whose handling combines detection, isolation and recovery. Intrusion-tolerance models in the same lineage, for instance the state-transition security model of Madan *et al.* [19], derive a mean time to security failure from a semi-Markov process with absorbing states. Our study lifts the reward-based and first-passage-based measures of such models from a single host to a footprint-coupled MEDC fleet with data-centric restore.

B. Edge micro-datacenters under attack

Liu *et al.* [13] model a lateral-movement attack on an MEDC as an absorbing CTMC with a uniform recovery rate into a good state, and the model serves as our direct antecedent and baseline. Roman *et al.* [3] survey the threat landscape of edge and fog deployments and motivate the exposure assumptions of our threat model. CRADLE differs from the antecedent in threat model and in objective. Our detection hazard is keyed on the encryption and exfiltration footprint rather than on the compromised-node count. Encryption becomes a recoverable denial state rather than a terminal breach. Our

response is a backup restore with recovery-point and recovery-time economics rather than a re-image.

C. Ransomware behaviour and defenses

Empirical and systems work dissects ransomware behaviour and the defenses against ransomware, covering file-system self-healing, real-time write monitoring and behavioural detection [4], [8], [9], [7]. Early comparative analyses of ransomware families document the cryptographic machinery underlying data denial [10], and the commoditized crimeware market explains the scale of modern campaigns [11]. Surveys of ransomware evolution record the shift toward double extortion, under which an operator exfiltrates data before encryption and threatens public release [6], [5]. Studies of advanced persistent threats describe the lateral-movement stages preceding the final payload [12]. The listed studies are measurement and mechanism studies, and no prior transient Markov reward model captures the exfiltrate-then-encrypt kill-chain in combination with detection and backup-restore economics.

D. Proactive recovery and storage dependability

Moving-target defense and rejuvenation-based recovery reset attacker progress on a timer or on a trigger [21], [22]. Our restore is conceptually a triggered reset, and the restore remains data-centric because the restored object is encrypted data drawn from an immutable backup and the residual cost is the recovery point. Storage dependability frameworks evaluate backup and mirroring designs against financial objectives [23], and CRADLE embeds a comparable cost-driven choice inside a live adversarial race.

E. Numerical methods

We use the matrix exponential for transient solutions, Jensen uniformization as an independent cross-check [24], a Van Loan style augmented exponential for accumulated rewards [25], [26], the fundamental matrix for per-incident expectations, phase-type arguments for the first-passage law [27], and the exact stochastic simulation algorithm of Gillespie for validation [28].

III. ARCHITECTURE, THREAT MODEL AND STOCHASTIC REWARD NET

A. Architecture

Fig. 1 depicts the deployment and the two defensive planes of the system. An MEDC hosts a fixed number of MEC servers behind one or more base stations, and each server runs a virtualization stack instantiating tenant services close to endpoint and cyber-physical devices. Two independent defensive planes overlay the compute fabric of the micro-datacenter. An EDR monitor continuously observes host and network telemetry, and the alert hazard of the monitor rises with the volume of ransomware-indicative activity such as high-entropy write bursts from encryption and anomalous data egress from exfiltration. An immutable-backup subsystem periodically snapshots server data to write-once storage. Upon an incident the subsystem drives an isolate-and-restore response, which network-isolates the MEDC and restores servers from the last good snapshot.

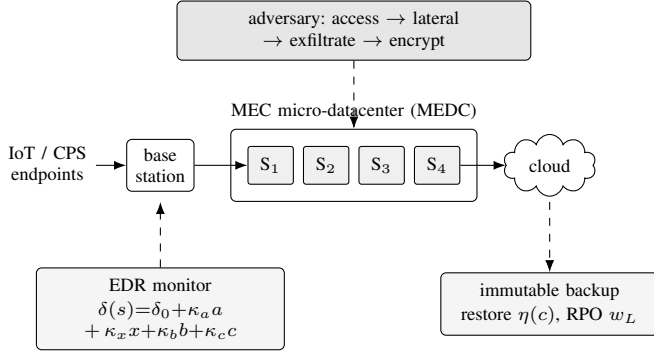


Fig. 1. CRADLE system architecture.

B. Threat model

The adversary follows a ransomware double-extortion kill-chain. The adversary obtains initial access to one server and then propagates laterally, and the propagation accelerates with the number of controlled servers. On a controlled server the adversary exfiltrates data, which creates confidentiality loss and extortion leverage, and only afterwards encrypts the same data until the server reaches a locked condition denying service. The defender remains unaware of the campaign until the EDR monitor fires, detection remains imperfect, and the timeliness of detection depends on the attacker footprint. We assume network isolation becomes effective once triggered, and isolation halts further spread and egress. Host-local encryption already in flight may complete before the end of the restore. Immutable backups are not corruptible, which is a design goal of modern write-once backup. We exclude the ransom-payment decision by design, because CRADLE is a defender-side resilience model built on a restore-not-pay premise.

C. State space

The MEDC holds a fixed number of servers, and every server occupies exactly one kill-chain stage. A transient operational state is the tuple

$$s = (a, x, b, c, d), \quad (1)$$

in which the first four entries count servers in the compromised, exfiltrating, encrypting and locked stages, and the last entry indexes the campaign phase with value zero for stealth and value one for response. The count of clean vulnerable servers, the campaign footprint and the active non-locked foothold follow from

$$p = a + x + b + c, \quad V = m - p, \quad h = a + x + b. \quad (2)$$

Validity requires non-negative counts and a footprint no larger than the fleet size. The stealth phase exists for every lattice point, and the response phase exists only for a non-empty footprint, because a campaign must exist before the start of a response. A single absorbing state denoted GOOD represents the fully restored and incident-resolved MEDC. Table I summarizes the encoding of every symbol and the corresponding

TABLE I
STATE ENCODING.

symbol	meaning	service
$V = m - p$	Vulnerable (clean, serving)	up
a	Compromised (foothold, staging)	up
x	Exfiltrating (data egress in progress)	up (leaking)
b	Encrypting (crypto in progress)	degraded
c	Locked (data denied)	down
d	phase: 0 stealth / 1 response	–
GOOD	restored (absorbing)	up

service condition. Counting the two phases gives the size of the transient state set,

$$|S| = 2 \binom{m+4}{4} - 1, \quad (3)$$

which amounts to 139 transient states plus one absorbing state for the base case of 4 servers.

D. Transition rules

Fig. 2 shows the kill-chain and the phase automaton, and the aggregate rate laws appear in Table II. In the stealth phase the adversary first obtains a foothold on a clean MEDC at the initial-access rate. A clean server afterwards becomes compromised through lateral movement at a rate proportional to the active foothold, which reproduces the epidemic footprint-scaling of the antecedent model [13]. Every compromised server begins exfiltration at a per-server onset rate. An exfiltrating server completes one stolen dataset and only afterwards begins encryption at a per-server completion rate. An encrypting server completes to the locked stage at a per-server encryption rate. The campaign becomes detected, and the phase index moves to the response value, at the footprint-aware hazard

$$\delta(s) = \delta_0 + \kappa_a a + \kappa_x x + \kappa_b b + \kappa_c c, \quad \delta(s) = 0 \text{ if } p=0, \quad (4)$$

whose stage gains obey the strict ordering

$$\kappa_b > \kappa_c > \kappa_x \geq \kappa_a. \quad (5)$$

An encrypting server sustains a high-entropy write storm and remains the loudest live signal, a locked server raises a user-visible outage alarm, an exfiltrating server leaves an egress signature, and a bare foothold stays nearly silent. In the response phase network isolation halts lateral movement, exfiltration and new encryption onsets. Host-local encryption already running completes at the same encryption rate, and the restore fires to GOOD at the state-dependent rate

$$\eta(c) = \frac{1}{\text{RTO}_0 + \text{RTO}_1 c}, \quad (6)$$

under which the recovery time grows with the number of locked servers awaiting restoration. With restore coverage in the unit interval, a complementary fraction of restores are incomplete and leave a residual foothold, and the campaign returns to a single compromised server.

exfiltrate-then-encrypt (double extortion)

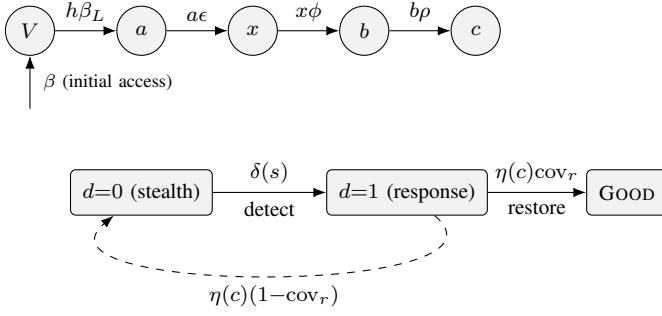


Fig. 2. Kill-chain stages and campaign-phase automaton.

TABLE II
TRANSITION RATE LAWS.

activity	effect	rate	guard
initial access	$(0, 0, 0, 0, 0) \rightarrow (1, 0, 0, 0, 0)$	β	$p=0, d=0$
lateral move	$a \rightarrow a+1$	$h\beta_L$	$V \geq 1, h \geq 1, d=0$
exfil. onset	$a \rightarrow a-1, x \rightarrow x+1$	$a\epsilon$	$a \geq 1, d=0$
exfil. done	$x \rightarrow x-1, b \rightarrow b+1$	$x\phi$	$x \geq 1, d=0$
encrypt done	$b \rightarrow b-1, c \rightarrow c+1$	$b\rho$	$b \geq 1$
detection	$d: 0 \rightarrow 1$	$\delta(s)$	$p \geq 1, d=0$
restore	$\rightarrow \text{GOOD}$	$\eta(c)\text{cov}_r$	$d=1$
partial restore	$\rightarrow (1, 0, 0, 0, 0)$	$\eta(c)(1-\text{cov}_r)$	$d=1$

E. Generator

The transient state set collects every operational state, and a single absorbing state completes the chain. The infinitesimal generator collects the arc rates through

$$q_{s,s'} = \sum_{\text{arcs } s \rightarrow s'} \lambda, \quad q_{s,s} = -\sum_{s' \neq s} q_{s,s'}, \quad (7)$$

which drives every row sum of the generator to zero. The row of the absorbing state contains only zero entries. Algorithm 1 synthesizes the generator from the rate laws, and a numerical assertion confirms the row sums. Default parameters per day appear in Table III, and the chosen values place the incident on a multi-day timescale in which detection, encryption and restore genuinely compete.

IV. MEASURES, SOLUTION AND DEFENDER CO-DESIGN

The initial distribution places all mass on the clean state, and the transient distribution obeys the matrix-exponential law

$$\pi(t) = \pi_0 e^{Qt}. \quad (8)$$

A. Transient and accumulated measures

The expected number of servers in a stage is a reward-weighted marginal of the transient distribution. The expected available service fraction and the probability of service denial both follow from the locked-server marginal,

$$\mathbb{E}[c(t)] = \sum_s c_s \pi_s(t), \quad A(t) = 1 - \frac{\mathbb{E}[c(t)]}{m}, \quad (9)$$

$$P_{\text{dn}}(t) = \mathbb{P}(c(t) \geq 1).$$

TABLE III
DEFAULT PARAMETERS.

symbol	meaning	default	unit
m	MEC servers in the MEDC	4	servers
β	initial-access rate	0.50	d^{-1}
β_L	lateral-movement rate / active foothold	0.40	d^{-1}
ϵ	exfiltration-onset rate / compromised	0.80	d^{-1}
ϕ	exfiltration-completion rate / exfiltrating	1.00	d^{-1}
ρ	encryption-completion rate / encrypting	2.00	d^{-1}
δ_0	baseline detection hazard	0.20	d^{-1}
κ_a	detection gain / compromised server	0.00	d^{-1}
κ_x	detection gain / exfiltrating server	0.50	d^{-1}
κ_b	detection gain / encrypting server	1.50	d^{-1}
κ_c	detection gain / locked server	0.80	d^{-1}
RTO_0	base restore time	0.25	d
RTO_1	restore-time growth / locked server	0.15	d
w_L	data loss / locked server (RPO)	1.00	unit
cov_r	restore coverage	1.00	—

Algorithm 1 CRADLE generator synthesis

Require: parameters θ , size m

- 1: enumerate $\mathcal{S} = \{(a, x, b, c, d) : a+x+b+c \leq m, d \in \{0, 1\}, (d=1 \Rightarrow p \geq 1)\}$
- 2: $Q \leftarrow 0$
- 3: **for all** $s = (a, x, b, c, d) \in \mathcal{S}$ **do**
- 4: **for all** (s', λ) in $\text{TRANSITIONS}(s; \theta)$ (Table II) **do**
- 5: $Q_{s,s'} \leftarrow Q_{s,s'} + \lambda, \quad Q_{s,s} \leftarrow Q_{s,s} - \lambda$
- 6: **end for**
- 7: **end for**
- 8: **assert** $\max_s |\sum_{s'} Q_{s,s'}| < \epsilon$
- 9: **return** Q

For a given reward-rate vector the expected accumulated reward to a chosen time horizon is obtained without quadrature from the augmented exponential

$$\begin{bmatrix} \pi(t) \\ \Lambda(t) \end{bmatrix} = \exp\left(\begin{bmatrix} Q^\top & 0 \\ I & 0 \end{bmatrix} t\right) \begin{bmatrix} \pi_0 \\ 0 \end{bmatrix}, \quad L_r(t) = r^\top \Lambda(t). \quad (10)$$

We instantiate the reward vector as the locked count for service-denial server-days, as the exfiltration-completion flux restricted to the stealth phase for stolen datasets, and as the restore flux weighted by the recovery point for data lost at restore.

B. First-passage distributions

A trajectory remains stealthy exactly until detection, and the resolution event coincides with absorption. Both first-passage distribution functions follow directly from the transient distribution,

$$\mathbb{P}(\text{detected} \leq t) = 1 - \sum_{s:d_s=0} \pi_s(t), \quad (11)$$

$$\mathbb{P}(\text{resolved} \leq t) = \pi_{\text{GOOD}}(t).$$

C. Per-incident expectations

Let the generator restricted to the transient states supply the fundamental matrix. The expected sojourn vector before absorption satisfies

$$\tau^\top = \pi_0^\top (-U)^{-1}, \quad (12)$$

and every per-incident total becomes a linear functional of the sojourn vector. The expected total time spent in stealth states gives the mean time to detection, the expected total time spent in response states gives the mean response duration, and the sum of both quantities gives the mean time to resolution. The same construction gives the expected number of exfiltrated datasets, the expected denial server-days, and the expected data loss at restore. The additive identity between the three time measures holds exactly, and a numerical check confirms the identity to machine precision.

D. Structural result

In a uniform-recovery model every transient state exits to the absorbing state at a single rate, and the resolution distribution function is exactly an exponential distribution function. CRADLE behaves in a qualitatively different manner, as the following statement makes precise.

Lemma 1 (Non-memoryless resolution). *Assume all rates of the generator are finite and assume the restore rate $\eta(\cdot)$ is finite. Then the time to resolution T_R , defined as the first passage from the clean state to GOOD, is a phase-type random variable whose density satisfies $f_{T_R}(0^+) = 0$. Consequently T_R is not exponentially distributed, and the resolution distribution function is not of the form $1 - e^{-\gamma t}$ for any $\gamma > 0$.*

Proof. Absorption into GOOD occurs only from response states, and response states are reachable only through the detection event, because no arc leads from a stealth state directly to GOOD. Every path to absorption therefore visits at least two distinct transient states, which gives $T_R \geq T_1 + T_2$ with T_1 the positive time to detection and T_2 the positive residual time to restore. As a first passage in a finite CTMC starting from a single transient state, T_R is phase-type [27]. At least two phases are traversed with probability one, hence the density at the origin vanishes and $f_{T_R}(0^+) = 0$. The exponential density equals $\gamma e^{-\gamma t}$ with a strictly positive value γ at the origin, and T_R cannot be exponential. $\square$

The stealth window imprints an initial dead time on the resolution curve, which produces the S shape. The coupling of detection to attack progress lies beyond the representational reach of a uniform-recovery model [13]. Section V quantifies the departure from the exponential law in numerical terms.

E. Security and dependability cost

We weight the incident harms into a single security-dependability loss and add the operational spend of the defender,

$$S = c_{br}\mathbb{E}[\text{exfil}] + c_{dn}\mathbb{E}[\text{denial}] + c_{dl}\mathbb{E}[\text{loss}] + c_{rt}\mathbb{E}[\text{response}], \quad (13)$$

$$J = S + \underbrace{C_{\text{mon}}(\sigma_D) + C_{\text{bkp}}(\tau_b)}_O. \quad (14)$$

in which the detection sensitivity and the backup cadence are the two defender levers introduced next.

Algorithm 2 Detection and backup co-design

Require: base parameters θ , cost config, lever boxes $[\sigma_{lo}, \sigma_{hi}], [\tau_{lo}, \tau_{hi}]$

- 1: evaluate J on a $\sigma_D \times \tau_b$ grid via Alg. 1 and (12), set $(\sigma_D, \tau_b) \leftarrow \text{grid minimizer}$
- 2: **repeat**
- 3: $\tau_b \leftarrow \text{GOLDENMIN}_{\tau} J(\sigma_D, \cdot)$
- 4: $\sigma_D \leftarrow \text{GOLDENMIN}_{\sigma} J(\cdot, \tau_b)$
- 5: **until** converged
- 6: **return** (σ_D^*, τ_b^*) , J^* , and the non-dominated frontier $\{(O, S)\}$

F. Solution and co-design

Algorithm 1 builds the generator from the rate laws of Table II. The transient distribution and the accumulated rewards follow from (10) by forming a single base-step propagator and composing the propagator along a uniform grid, which introduces no step error. The per-incident totals follow from the single linear solve of (12). We cross-check the matrix exponential against Jensen uniformization at every grid point.

The defender holds two orthogonal investment levers over the whole design space. Detection sensitivity scales the three non-zero footprint coefficients of (4) by a common multiplier no smaller than one. A sharper EDR monitor shortens the stealth window and costs more to run, and the monitoring cost grows linearly in the multiplier. The backup cadence sets the recovery point of the restore in hours. The data loss per locked server scales linearly with the cadence, and the backup and storage overhead scales with the backup frequency. Algorithm 2 minimizes the expected total cost of (14) over both levers. The algorithm maps the surface on a grid and then polishes the solution with alternating one-dimensional golden-section searches, and each objective evaluation is one solve of (12). Sweeping the levers and retaining non-dominated pairs of spend and loss traces the Pareto frontier.

G. Independent simulation

Algorithm-agnostic validation comes from an exact Gillespie simulation [28] consuming only the per-state transition list. The simulator evolves each trajectory from the clean condition to the restored condition, and the simulator rebuilds the transient probabilities, the distribution functions and the per-incident totals with confidence intervals.

V. VALIDATION AND NUMERICAL RESULTS

Unless stated otherwise, results use the defaults of Table III with 4 servers over a horizon of 20 days. Analytic curves are lines, and independent simulation estimates are filled markers.

A. Verification and validation

We validate the analytic engine of the study in three independent ways. The matrix exponential agrees with Jensen uniformization to solver tolerance near 10^{-12} . The fundamental-matrix per-incident totals agree with the time-integrated accumulated reward of (10) to a residual near 10^{-15} . An independent Gillespie simulation of 200,000 trajectories reproduces

TABLE IV
ANALYTIC AND SIMULATION AGREEMENT.

quantity	analytic	simulation (95% CI)	agree
E[datasets exfiltrated]	0.5657	0.5642 ± 0.0028	✓
E[denial] (srv-days)	0.3894	0.3854 ± 0.0037	✓
E[data loss] (RPO)	0.3823	0.3801 ± 0.0025	✓
MTTD (days)	3.6366	3.6308 ± 0.0101	✓
MTTResolve (days)	3.9440	3.9369 ± 0.0103	✓

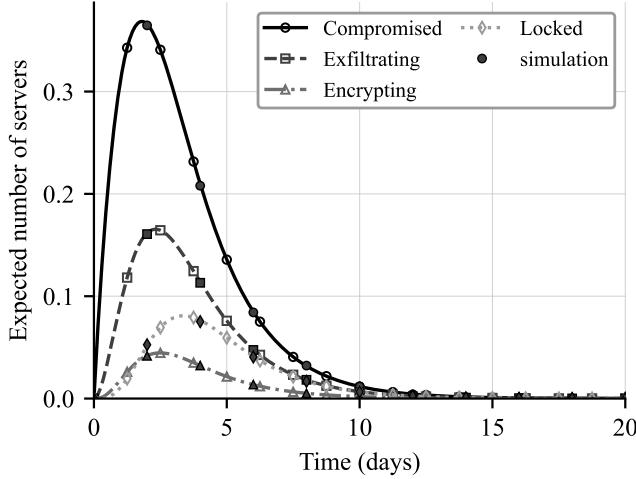


Fig. 3. Kill-chain stage populations.

every analytic quantity. Table IV records the third leg, in which every per-incident expectation agrees within the 95% confidence interval. The detection and resolution distribution functions agree with a maximum discrepancy below 0.01. All seventeen automated checks pass.

B. Kill-chain dynamics and service denial

Fig. 3 tracks the expected number of servers in each kill-chain stage. The compromise and exfiltration populations rise first and feed a delayed wave of encryption and of locking. All populations then collapse once detection triggers isolation and restore. Fig. 4 translates the stage dynamics into service availability and service denial terms. The expected available fraction dips to 98.0% at day 3.4 during the encryption wave, and the probability of service denial peaks at 7.5% near day 3.2 before restoration returns the MEDC to full service. The transient nature of the incident is essential, because a steady-state model misses the peak denial entirely.

C. Accumulated security and dependability loss

Fig. 5 decomposes the accumulated expected loss into exfiltration, service denial, recovery-point data loss and response components. The loss accrues steeply during the active phase and saturates after incident resolution. The composition across confidentiality, availability and recency tells a defender which harm dominates and hence which lever deserves investment.

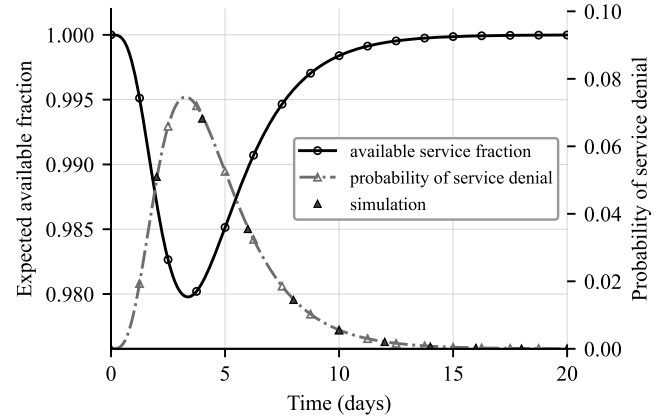


Fig. 4. Service availability and denial probability.

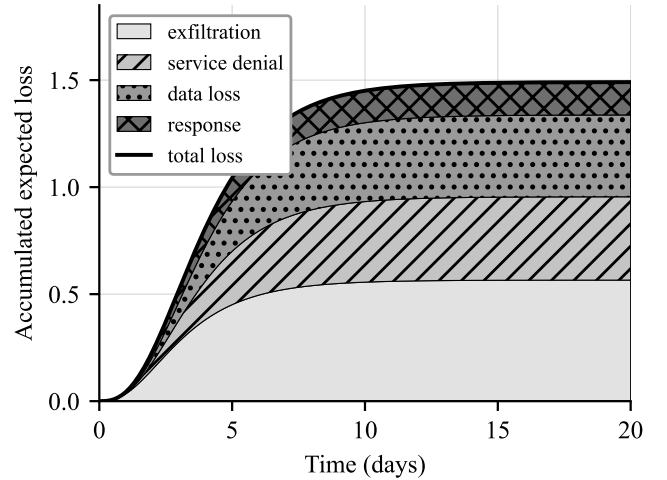


Fig. 5. Accumulated loss decomposition.

D. Detection race and non-memoryless recovery

Fig. 6 shows the detection distribution function for a range of detection sensitivities, including one degraded setting below the lower bound of the co-design lever box. Sharper detection compresses the stealth window, and the baseline hazard keeps the detection probability rising toward one even for a dull monitor, because every campaign eventually generates a footprint. Fig. 7 carries the headline structural result of the whole study. The resolution curve takes a clear S shape with an initial dead time while the campaign remains stealthy, and the curve departs from the best-fit memoryless exponential by 0.21 at the maximum. A uniform-recovery model of the form of [13] cannot represent such behaviour, and the coupling of detection to attack progress makes edge ransomware resolution non-memoryless.

E. Defender co-design

Fig. 8 maps the expected total cost over the plane of detection sensitivity and backup cadence, and a fine grid sweep locates a single interior basin. Cost rises steeply at short backup intervals through storage overhead and rises more

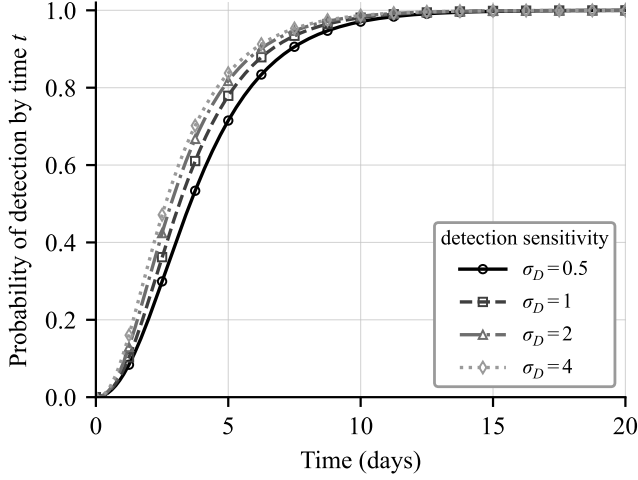


Fig. 6. Detection distribution function.

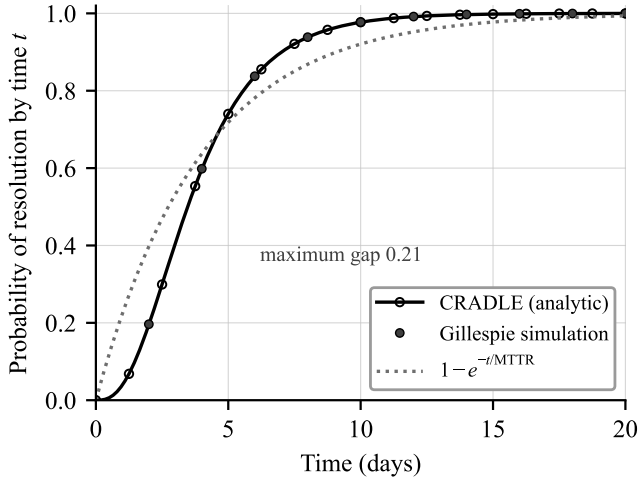


Fig. 7. Resolution probability against exponential fit.

gently at high sensitivity through monitoring overhead, while very loose backups or a dull monitor raise the security loss well above the optimum. Algorithm 2 locates the optimum at a sensitivity of 2.17 and a backup interval of 32 hours, where the security loss and the operational spend end up nearly balanced. Fig. 9 traces the corresponding Pareto frontier, and the minimum-total-cost design sits at the point where the marginal security gain equals the marginal spend. Table V contrasts the baseline, a detection-blind backup-only optimizer, and the co-design.

The counterintuitive finding lies in the levers themselves (Figs. 10 and 11). A detection-blind operator cannot reduce the number of servers reaching the locked stage, and such an operator therefore spends more of the budget on frequent backups and nothing extra on detection, which costs a penalty of 14% in total cost relative to the co-design. The co-designer instead sharpens detection until far fewer servers ever reach the locked stage, which cuts expected denial by 62% and expected data loss by 23%. The co-designer can then

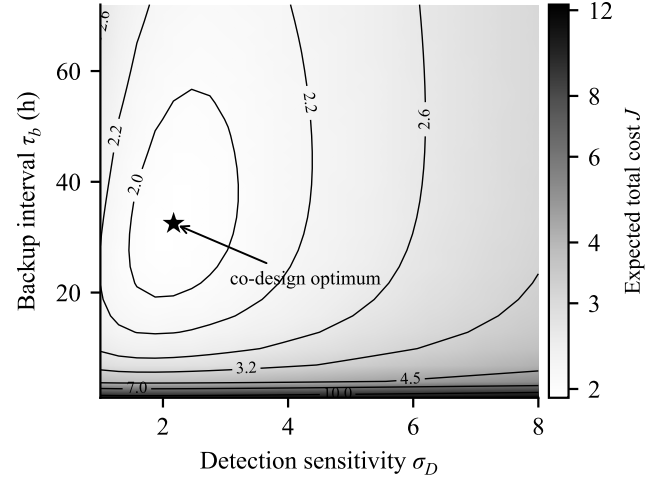


Fig. 8. Expected total cost surface.

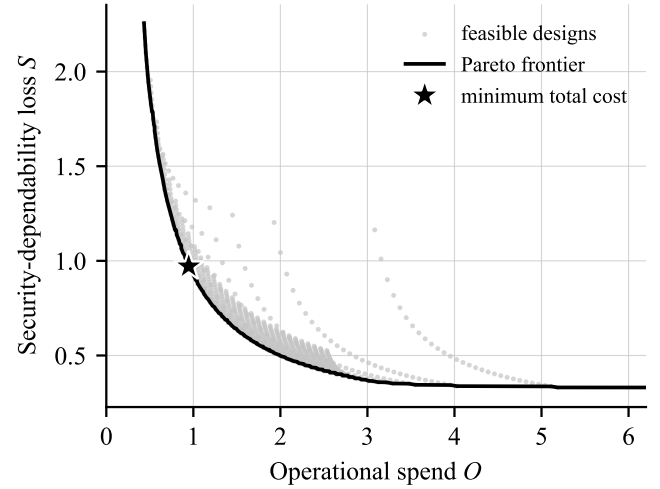


Fig. 9. Security-loss and operational-spend frontier.

relax the backup interval and save storage overhead. Better detection buys looser backups, because the two levers act as economic substitutes, and tuning the levers independently remains strictly sub-optimal.

F. Fleet size and parameter sensitivity

Fig. 12 scales the MEDC from 2 to 6 servers, which grows the state space from 29 to 419 transient states. The per-incident harms saturate, because detection resolves the incident before the campaign traverses the whole fleet, and the expected number of exfiltrated datasets and the denial server-days barely grow with fleet size. For edge micro-datacenters the practical message names detection latency and backup recency as the governing factors of ransomware risk. Fig. 13 ranks the drivers of expected data loss by scaled sensitivity. The backup recovery point dominates with unit elasticity, the exfiltration-completion rate and the encryption-completion rate follow, and every detection coefficient reduces the loss,

TABLE V
BASELINE, DETECTION-BLIND AND CO-DESIGN COMPARISON.

quantity	baseline	detection-blind	co-design
detection sensitivity σ_D	1.00	1.00	2.17
backup interval τ_b (h)	24	25	32
datasets exfiltrated	0.566	0.566	0.385
denial (server-days)	0.389	0.389	0.149
data loss (RPO units)	0.382	0.391	0.296
security loss S	1.491	1.500	0.971
operational spend O	0.700	0.691	0.947
total cost J	2.191	2.191	1.918

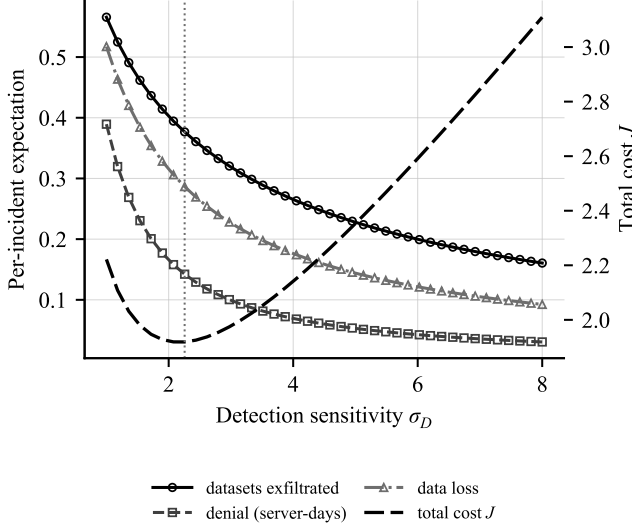


Fig. 10. Effect of detection sensitivity.

which confirms the co-design intuition. The compromised-stage detection gain is zero at the baseline and admits no scaled sensitivity, hence the ranking omits the coefficient.

G. Robustness to imperfect restore

Real restores sometimes fail to evict the adversary completely, because a restored server can be re-infected from a missed persistence foothold, and double-extortion crews often re-encrypt in order to punish incomplete recovery. Fig. 14 sweeps the restore coverage from a perfect value down to 0.6. As coverage degrades, a fraction of incidents relapse into a fresh campaign before final resolution, and both the mean time to resolution and the expected data loss inflate in a convex manner. The inflation reaches 21% in resolution time and 43% in data loss at a coverage of 0.7. The practical corollary elevates restore quality, meaning a verified, immutable and fully evicting restore, into a first-order concern, because a cheap and leaky restore erases the benefit of an otherwise well-tuned co-design.

H. Design guidance and threats to validity

Three guidelines follow from the numerical study. First, instrumenting for the crypto footprint pays best, because the detection hazard rises with encryption input-output and

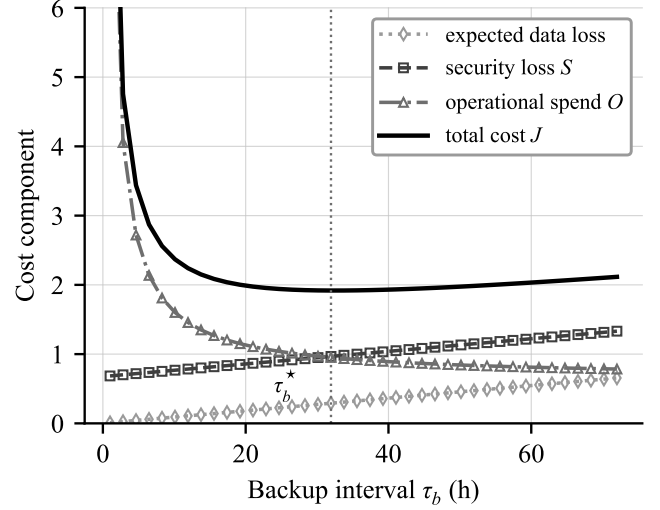


Fig. 11. Effect of backup interval.

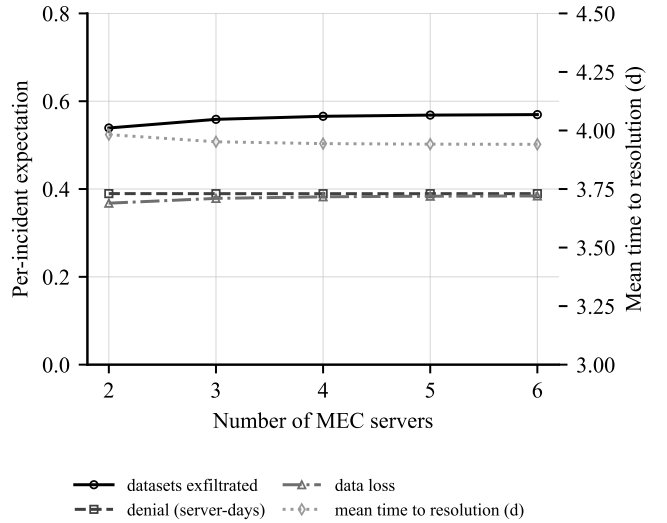


Fig. 12. Per-incident measures against fleet size.

tuning the monitor toward high-entropy write bursts gives the steepest security return. Second, detection and backups deserve joint budgeting, because the two levers act as substitutes and separate budgeting causes over-spending. Third, fleet size is not the dominant risk factor, because per-incident harm saturates and consolidation for efficiency does not materially raise ransomware exposure once detection and restore are in place.

Several modelling assumptions bound the scope of the present study. CRADLE assumes exponential stage sojourns, and heavy-tailed encryption or restore times need a semi-Markov or phase-type refinement, which the accumulated-reward and fundamental-matrix machinery accommodates directly. We assume incorruptible immutable backups and effective isolation, and relaxing the two assumptions adds arcs without new formalism. The cost weights are illustrative, the

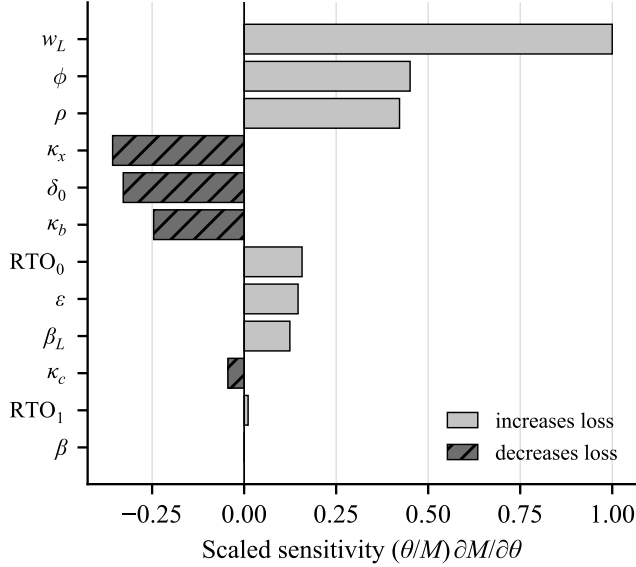


Fig. 13. Scaled sensitivity of expected data loss.

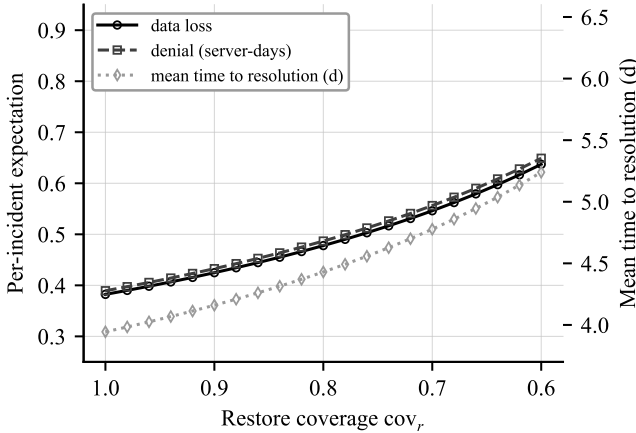


Fig. 14. Robustness to imperfect restore coverage.

framework consumes any weights, and the qualitative guidance on lever substitutability and on saturation with fleet size stayed stable across the sweeps of the study. Finally, the ransom-payment game falls outside the scope by design, because CRADLE models the resilience of a restore-not-pay defender.

VI. CONCLUSION

The study presented CRADLE, a transient stochastic-reward-net model of a crypto-ransomware double-extortion kill-chain on a MEC micro-datacenter, defended by a footprint-aware endpoint monitor and by an immutable-backup isolate-and-restore response. Unlike prior edge-under-attack models, CRADLE treats encryption as a recoverable service-denial state rather than an absorbing breach, orders exfiltration before encryption in order to capture double extortion, ties the detection hazard to the encryption and exfiltration footprint, and endows the response with explicit recovery-point and recovery-time economics. From one generator we obtained

transient stage and service-denial probabilities, accumulated-reward losses in confidentiality, availability and data recency, first-passage detection and resolution distributions, and closed per-incident expectations through the fundamental matrix. We proved and demonstrated a qualitative departure from uniform-recovery models, because coupling detection to attack progress makes the resolution law non-memoryless. A co-design procedure tuned detection sensitivity and backup cadence jointly to minimize expected total cost, revealed a Pareto frontier and the substitutability of the two levers, under which sharper detection lets a defender back up less often, while per-incident harm saturates with fleet size. An independent exact simulation validated every analytic result of the whole study. Future work lifts the exponential assumption toward semi-Markov sojourns for heavy-tailed encryption and restore times, embeds an adaptive state-triggered response policy, extends the model to federations of interacting micro-datacenters, and incorporates the strategic ransom-payment decision as a game against the modeled defender.

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